

Q1 - 24 June - Shift 1

The number of choices of $\Delta \in \{\wedge, \vee, \Rightarrow, \Leftrightarrow\}$, such that $(p\Delta q) \Rightarrow ((p\Delta \sim q) \vee ((\sim p)\Delta q))$ is a tautology, is

- (A) 1 (B) 2
(C) 3 (D) 4

*Space for your notes:***Q2 - 24 June - Shift 2**

Consider the following statements :

A : Rishi is a judge.

B : Rishi is honest.

C : Rishi is not arrogant.

The negation of the statement "if Rishi is a judge and he is not arrogant, then he is honest" is

- (A) $B \rightarrow (A \vee C)$
(B) $(\sim B) \wedge (A \wedge C)$
(C) $B \rightarrow ((\sim A) \vee (\sim C))$
(D) $B \rightarrow (A \wedge C)$

*Space for your notes:***Q3 - 25 June - Shift 1**

Questions

MathonGo

Consider the following two propositions:

$$P1 : \sim(p \rightarrow \sim q)$$

$$P2 : (p \wedge \sim q) \wedge ((\sim p) \vee q)$$

If the proposition $p \rightarrow ((\sim p) \vee q)$ is evaluated as FALSE, then:

- (A) P1 is TRUE and P2 is FALSE
- (B) P1 is FALSE and P2 is TRUE
- (C) Both P1 and P2 are FALSE
- (D) Both P1 and P2 are TRUE

Space for your notes:

Q4 - 25 June - Shift 2

The negation of the Boolean expression $((\sim q) \wedge p) \Rightarrow ((\sim p) \vee q)$ is logically equivalent to

- (A) $p \Rightarrow q$
- (B) $q \Rightarrow p$
- (C) $\sim(p \Rightarrow q)$
- (D) $\sim(q \Rightarrow p)$

Space for your notes:

Q5 - 26 June - Shift 1

Let $\Delta, \nabla \in \{\wedge, \vee\}$ be such that $p \nabla q \Rightarrow ((p \nabla q) \Delta r)$ is a tautology. Then $(p \nabla q) \Delta r$ is logically equivalent to :

- (A) $(p \Delta r) \vee q$
- (B) $(p \Delta r) \wedge q$
- (C) $(p \wedge r) \Delta q$
- (D) $(p \nabla r) \wedge q$

Space for your notes:

Q6 - 26 June - Shift 2

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Questions

MathonGo

Let $r \in \{p, q, \sim p, \sim q\}$ be such that the logical statement $r \vee (\sim p) \Rightarrow (p \wedge q) \vee r$ is a tautology.

Then 'r' is equal to :

- (A) p (B) q (C) $\sim p$ (D) $\sim q$

Space for your notes:

Q7 - 27 June - Shift 1

The Boolean expression $(\sim(p \wedge q)) \vee q$ is equivalent to :

- (A) $q \rightarrow (p \wedge q)$ (B) $p \rightarrow q$
(C) $p \rightarrow (p \rightarrow q)$ (D) $p \rightarrow (p \vee q)$

Space for your notes:

Q8 - 27 June - Shift 2

Which of the following statement is a tautology?

- (A) $((\sim q) \wedge p) \wedge q$
(B) $((\sim q) \wedge p) \wedge (p \wedge (\sim p))$
(C) $((\sim q) \wedge p) \vee (p \vee (\sim p))$
(D) $(p \wedge q) \wedge (\sim(p \wedge q))$

Space for your notes:

Q9 - 28 June - Shift 1

Questions

MathonGo

Let p, q, r be three logical statements. Consider the compound statements

$$S_1 : ((\sim p) \vee q) \vee ((\sim p) \vee r) \quad \text{and}$$

$$S_2 : p \rightarrow (q \vee r)$$

Then, which of the following is **NOT** true ?

(A) If S_2 is True, then S_1 is True

(B) If S_2 is False, then S_1 is False

(C) If S_2 is False, then S_1 is True

(D) If S_1 is False, then S_2 is False

*Space for your notes:***Q10 - 28 June - Shift 2**

Suppose a class has 7 students. The average marks of these students in the mathematics examination is 62, and their variance is 20. A student fails in the examination if he/she gets less than 50 marks, then in worst case, the number of students can fail is

*Space for your notes:***Q11 - 28 June - Shift 2**

The maximum number of compound propositions, out of $p \vee r \vee s$, $p \vee r \vee \sim s$, $p \vee \sim q \vee s$, $\sim p \vee \sim r \vee s$, $\sim p \vee \sim r \vee \sim s$, $\sim p \vee q \vee \sim s$, $q \vee r \vee \sim s$, $q \vee \sim r \vee \sim s$, $\sim p \vee \sim q \vee \sim s$ that can be made simultaneously true by an assignment of the truth values to p, q, r and s , is equal to

*Space for your notes:***Q12 - 29 June - Shift 1**

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Questions

MathonGo

Let $\Delta \in \{\wedge, \vee, \Rightarrow, \Leftrightarrow\}$ be such that

Space for your notes:

$(p \wedge q) \Delta ((p \vee q) \Rightarrow q)$ is a tautology. Then Δ is equal to :

(A) $\wedge$

(B) $\vee$

(C) $\Rightarrow$

(D) $\Leftrightarrow$

Q13 - 29 June - Shift 2

Negation of the Boolean statement

Space for your notes:

$(p \vee q) \Rightarrow ((\sim r) \vee p)$ is equivalent to:

(A) $p \wedge (\sim q) \wedge r$

(B) $(\sim p) \wedge (\sim q) \wedge r$

(C) $(\sim p) \wedge q \wedge r$

(D) $p \wedge q \wedge (\sim r)$

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Answer Key

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Q1 (B)

Q2 (B)

Q3 (C)

Q4 (C)

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Q5 (A)

Q6 (C)

Q7 (D)

Q8 (C)

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Q9 (C)

Q10 (0)

Q11 (9)

Q12 (C)

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Q13 (C)

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Q1 (B)

For tautology $((p \Delta \sim q) \vee ((\sim p) \Delta q))$ must be true.

This is possible only when $\Delta = \vee \& \Rightarrow$

Q2 (B)

$$\sim((A \wedge C) \rightarrow B)$$

$$\sim(\sim(A \wedge C) \vee B)$$

Using De-Morgan's law

$$(A \wedge C) \wedge (\sim B)$$

Option B is correct.

Q3 (C)

p	q	$\sim p$	$\sim q$	$\sim p \vee q$	$p \rightarrow (\sim p \vee q)$	$p \rightarrow \sim q$	$\sim(p \rightarrow \sim q)$	$p \wedge \sim q$	p
T	T	F	F	T	T	F	T	F	T
T	F	F	T	F	F	T	F	T	F
F	T	T	F	T	T	F	F	F	T
F	F	T	T	T	T	T	F	F	F

$p \rightarrow (\sim p \vee q)$ is F when p is true q is false

From table

P1 & P2 both are false

Q4 (C)

$$\sim p \vee q \equiv p \rightarrow q$$

$$\sim q \wedge p \equiv \sim(p \rightarrow q)$$

Negation of $\sim(p \rightarrow q) \rightarrow (p \rightarrow q)$

is $\sim(p \rightarrow q) \wedge (\sim(p \rightarrow q))$ i.e. $\sim(p \rightarrow q)$

Q5 (A)**Case-I** If $\Delta \equiv \nabla \equiv \wedge$

$$(p \wedge q) \rightarrow ((p \wedge q) \wedge r)$$

it can be false if r is false,

so not a tautology

Case-II If $\Delta \equiv \nabla \equiv \vee$

$$(p \vee q) \rightarrow ((p \vee q) \vee r) \equiv \text{tautology}$$

$$\text{then } (p \vee q) \vee r \equiv (p \Delta r) \vee q$$

Case-III if $\Delta = \vee, \nabla = \wedge$

$$\text{then } (p \wedge q) \rightarrow \{(p \vee q) \wedge r\}$$

Not a tautology

(Check $p \rightarrow T, q \rightarrow T, r \rightarrow F$)**Case-IV** if $\Delta = \wedge, \nabla = \vee$

$$(p \wedge q) \rightarrow \{(p \wedge q) \vee r\}$$

Not a tautology

Q6 (C)

By options

(1)

$p \Rightarrow r$	q	$\sim p$	$r \vee (\sim p)$	$(p \wedge q)$	$(p \wedge q) \vee r$	$r \vee (\sim p) \Rightarrow (p \wedge q) \vee r$
T	F	F	T	F	T	T
F	T	T	T	F	F	F
T	T	F	T	T	T	T
F	F	T	T	F	F	T

(2)

p	$\sim p$	$r \vee (\sim p)$	$q \Rightarrow r$	$(p \wedge q)$	$(p \wedge q) \vee r$	$r \vee (\sim p) \Rightarrow (p \wedge q) \vee r$
T	F	T	T	T	T	T
F	T	T	T	F	T	T
T	F	F	F	F	F	T
F	T	T	F	F	F	F

(3)

p	q	$r = \sim p$	$r \vee (\sim p)$	$(p \wedge q)$	$(p \wedge q) \vee r$	$r \vee (\sim p) \Rightarrow (p \wedge q) \vee r$
T	T	F	F	T	T	T
F	T	T	T	F	T	T
T	F	F	F	F	F	T
F	F	T	T	F	T	T

(4)

$\sim p$	p	q	$r \vee (\sim p)$	$r = \sim q$	$(p \wedge q)$	$(p \wedge q) \vee r$	$r \vee (\sim p) \Rightarrow (p \wedge q) \vee r$
F	T	T	F	F	T	T	T
F	T	F	T	T	F	T	T
T	F	T	T	F	F	F	F
T	F	F	T	T	F	T	T

Now final answer is option no. 3.

Q7 (D)

$$(\sim(p \wedge q)) \vee q$$

$$= (\sim p \vee \sim q) \vee q$$

$$= \sim p \vee \sim q \vee q$$

$$= \sim p \vee t$$

= this statement is a tautology option D

$p \Rightarrow (p \vee q)$ is also a tautology.

OR

p	q	$P \wedge q$	$\sim(p \wedge q)$	$\sim(p \wedge q) \vee q$	$p \vee q$	$p \rightarrow (p \vee q)$
T	T	T	F	T	T	T
T	F	F	T	T	T	T
F	T	F	T	T	T	T
F	F	F	T	T	F	T

Q8 (C)

$$(A) (\sim q \wedge p) \wedge q = (\sim q \wedge q) \wedge p = f$$

$$(B) (\sim q \wedge p) \wedge (p \wedge \sim p) = \sim q \wedge (p \wedge \sim p) = f$$

$$(C) (\sim q \wedge p) \vee (p \vee \sim p) = (\sim q \wedge p) \vee (t) = t$$

$$(D) (p \wedge q) \wedge (\sim(p \wedge q)) = f$$

Q9 (C)

$$s_1 : (\sim p \vee q) \vee (\sim p \vee r)$$

$$\equiv \sim p \vee (q \vee r)$$

$$s_2 : p \rightarrow (q \vee r)$$

$$\equiv \sim p \vee (q \vee r) \rightarrow \text{By conditional law}$$

$$s_1 \equiv s_2$$

Q10 (0)

Hints and Solutions

MathonGo

$$20 = \frac{\sum_{i=1}^7 |x_i - 62|^2}{7}$$

$$\Rightarrow |x_1 - 62|^2 + |x_2 - 62|^2 + \dots + |x_7 - 62|^2 = 140$$

If $x_1 = 49$

$$|49 - 62|^2 = 169$$

then,

$$|x_2 - 62|^2 + \dots + |x_7 - 62|^2 = \text{Negative Number}$$

which is not possible, therefore, no student can fail.

Q11 (9)

If we take

p	q	r	s
F	F	T	F

The truth value of all the propositions will be true.

Q12 (C)

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$$p \vee q \Rightarrow q$$

$$\Rightarrow \sim(p \vee q) \vee q$$

$$\Rightarrow (\sim p \wedge \sim q) \vee q$$

$$\Rightarrow (\sim p \vee q) \wedge (\sim q \vee q)$$

$$\Rightarrow (\sim p \vee q) \wedge t = \sim p \vee q$$

Now by taking option C

$$(p \wedge q) \Rightarrow \sim p \vee q$$

$$\Rightarrow \sim p \vee \sim q \vee \sim p \vee q$$

$$\Rightarrow t$$

Hence C

Q13 (C)

$$P \vee q \Rightarrow (\sim r \vee p)$$

$$\equiv \sim(p \vee q) \vee (\sim r \vee p)$$

$$\equiv (\sim p \wedge \sim q) \vee (p \vee \sim r)$$

$$\equiv [\sim p \vee p) \wedge (\sim q \vee p)] \vee \sim r$$

$$\equiv [\sim q \vee p) \vee \sim r$$

Its negation is $\sim p \wedge q \wedge r$